Capacity-Constrained CVRP Clustering on an Analog Neutral-Atom Register

Track A Report — DuQuantum Hackathon — Team Qualies

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Abstract

This work investigates a cluster-first decomposition of the capacitated vehicle-routing problem (CVRP) for an analog neutral-atom processor. Customer-to-vehicle assignment is encoded with one atom per binary decision, after which each selected cluster is routed exactly. The resulting formulation requires 8, 30, and 60 atoms on tiers S, M, and L, respectively. Raw quantum sampling is reliable on tier S, substantially less accurate on tier M, and rarely feasible on tier L. A held-out capacity study shows that a stronger load-balance coefficient can improve average feasibility, albeit with instance-dependent degradation in route quality. A depot-aware classical clustering surrogate substantially reduces the decomposition loss on tier L, whereas deterministic repair provides no advantage over identically processed random one-hot controls. These findings distinguish errors arising from the clustering objective, capacity penalization, geometric embedding, and quantum sampling.

1. Mathematical formulation

Formulation overview. The register determines which customers share a vehicle, and each resulting cluster is routed exactly. One atom is assigned to each customer–vehicle pair.

Rationale for cluster-first decomposition. Encoding the full route (customer, vehicle, position) requires approximately N^2K atoms: 32 / 300 / 900 for S / M / L, against 80 available (DERIVATIONS Q1). Deciding only the grouping needs NK. The cost is a *decomposition gap*: the best grouping under our objective is not always the grouping of the shortest routes. It is 0 % on 5 of 11 instances and up to 21.7 % on the rest (§5, decomposition table).

Decision variables. Let N denote the number of customers with demands d_i, K the number of identical vehicles of capacity Q, and d_{ij} the distance between customers i and j. The bit $x_{ik} = 1$ means customer i rides on vehicle k. A valid grouping has exactly one 1 in every customer's row ("one-hot").

Clustering objective. Spatially compact groups are encouraged by summing the distances between every pair of customers that share a vehicle:

$$f(x) = \sum_{k=1}^{K} \sum_{i<j} d_{ij}\, x_{ik}\, x_{jk}. \tag{1}$$

This surrogate introduces the decomposition gap reported below. Its additivity permits exact certification of the model optimum by dynamic programming (`cvrp/exact.py`).

Assignment constraint. The requirement that every customer be assigned to exactly one vehicle is enforced by a squared penalty with weight A:

$$A \sum_{i=1}^{N} \Big(1 - \sum_{k=1}^{K} x_{ik}\Big)^2. \tag{2}$$

Expansion produces a uniform linear term $-A$ and coupling $+2A$ between the K atoms of each customer. No additional atoms are required, although A becomes the largest coupling that the register must realize.

Capacity constraint. For vehicle load $L_k = \sum_i d_i x_{ik} \le Q$, the formulation uses unbalanced penalization (Montañez-Barrera et al. 2024), $\sum_k \big[-\lambda_1(Q - L_k) + \lambda_2(Q - L_k)^2\big]$, because it needs no slack atoms. On one-hot strings its linear part is the same constant for every valid grouping, so it drops out together with λ_1 (the "one-hot identity", DERIVATIONS §4); what survives is a load-balance term

$$\lambda_2 \sum_{k=1}^{K} L_k^2 \;=\; \text{const} + 2\lambda_2 \sum_{k} \sum_{i<j} d_i d_j\, x_{ik} x_{jk}. \tag{3}$$

The balance term *alone* has a feasible minimiser for $K = 2$ whenever a feasible split exists; adding the distance objective removes that guarantee, and no such guarantee exists for $K \ge 3$. We therefore score feasible decodes only.

Distinction among optimization objectives. The constrained problem minimizes Eq. (1) subject to both constraints. The QUBO sent to the embedder is, up to the offset AN,

$$E(x) = -A \sum_{ik} x_{ik} + 2A \sum_{i} \sum_{k<\ell} x_{ik} x_{i\ell} + \sum_{k} \sum_{i<j} (d_{ij} + 2\lambda_2 d_i d_j) x_{ik} x_{jk}. \tag{4}$$

The device realises $E_{\text{dev}}(x) = -\delta_f \sum_a x_a + \sum_{a<b} U_{ab} x_a x_b$. Positive couplings and uniform linear terms make the QUBO compatible in sign, but planar $U_{ab} = C_6/r_{ab}^6$ still approximates it.

Penalty coefficients. $A = \max_i \sum_{j\neq i}(d_{ij} + 2\lambda_2 d_i d_j) + 1$, which guarantees that every minimizer is one-hot (DERIVATIONS §4). $\lambda_2 = 0.15\, R/(2Q)$ with $R = \max_i \sum_j d_{ij}$, from a 40-seed brute-force scan. Brute force on the five S instances: the minimiser is feasible and optimal on 5/5; A survives one halving and fails at $A/4$; λ_2 is not binding on any S instance.

2. Resource requirements and instances

Encoding	count	S	M	L	fits 80?
assignment variables only	NK	8	30	60	yes
+ binary slack (textbook)	$NK + K\lceil \log_2(Q{+}1) \rceil$	14	45	80	yes (L = cap)
ours (`onehot-ub`)	NK	**8**	**30**	**60**	yes

Tiers: S $N{=}4, K{=}2, Q{=}7$; M $N{=}10, K{=}3, Q{=}20$; L $N{=}15, K{=}4, Q{=}30$ (`scripts/variable_count.py`). We save 6 / 15 / 20 atoms over the textbook encoding and never discard a readout bit.

Instances. A seeded generator first draws a balanced assignment and then draws demands within each vehicle's remaining capacity; that assignment is the shipped certificate and the validator rechecks it. Locations are uniform in $[0, 100]^2$ and the depot is fixed at $(50, 50)$. Across S/M/L, ρ ranges are 0.643–0.857 / 0.700–0.950 / 0.600–0.708 and $d_{\max}/Q$ ranges are 0.429–0.571 / 0.400 / 0.300–0.333.

3. Geometric embedding (DERIVATIONS §5)

Embedding strategy. A generic least-squares placement does not realize the rook's graph with sufficient accuracy. The adopted method therefore fits device energies on one-hot strings. On the five tested S instances the realised register preserves the optimal clustering; it does not reproduce every model energy exactly. On M/L it fails to preserve the ordering of groupings.

Limitations of generic placement. The K atoms representing one customer must be separated by 5 µm to form the required clique. Any atom of another customer is then at nearly the same distance from all K of them, so the couplings that should be zero (different customer *and* different vehicle) come out as large as the wanted ones. Measured: relative coupling error 0.95, defined as $\|U/\alpha - J\|_F / \|J\|_F$ over all pairs (lower is better; `cvrp/embed.py`), and on S_seed04 the register's ground state was an infeasible string.

One-hot-subspace fitting. On a one-hot string, each pair of customers has exactly one active coupling; consequently, only the difference *same-vehicle minus cross-vehicle* has to match the model (DERIVATIONS 5.7b); the cross couplings only need to be constant. Each customer becomes a rigid 5 µm K-gon, which realises the clique exactly with $\alpha = C_6/(5\,\mu\text{m})^6/2A$; the centres and orientations are fitted (L-BFGS, 6 seeded starts) so that the device energies of all one-hot strings match $\alpha\, E_{\text{QUBO}}$, with single-flip defects forced above their parents and the final detuning δ_f as a fit parameter. Registers are committed in `embeddings/`; every run checks the realised register (is its own minimiser feasible and optimal? smallest defect gap). Tier S: 5/5 yes, coupling error 0.21–0.32.

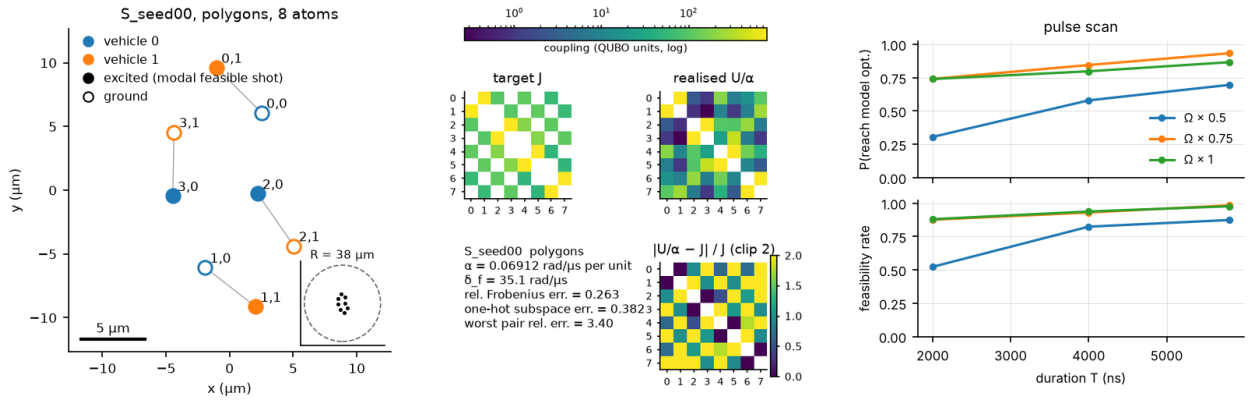


Figure 1: Left: S_seed00 register and modal feasible sample. Middle: target and realized couplings; the one-hot-subspace error is 0.382. Right: tier-S pulse scan; 0.75 times maximum amplitude at 5800 ns performs best among the tested settings.

An isolated-polygon bound. Ignoring other polygons, one atom of a K-gon excited costs $-\delta_f$; two excited cost $-2\delta_f + U_{\min}$, where $U_{\min}$ is the coupling of the *weakest* pair inside the polygon. One one excitation is preferred only if

$$0 < \delta_f < U_{\min}. \tag{5}$$

For $K \le 3$ every pair is the 5 µm side, $U_{\min} = 55.1$ rad/µs; the S fits land at $\delta_f = 35$–38 rad/µs (0.64–0.68 $U_{\min}$). On M the fit is ill-conditioned and ran to the device maximum, 125.66 rad/µs $= 2.28\,U_{\min}$, which favours *two* vehicles per customer; the embedder now enforces Eq. (4) (fallback $0.65\,U_{\min}$). For $K = 4$ a 5 µm square has 7.07 µm diagonals, and $1/r^6$ turns that $\sqrt{2}$ into a factor 8: $U_{\min} = 6.9$ rad/µs, so the L registers run at $\delta_f = 4.5$ rad/µs. Four points in a plane are never equidistant, so a planar one-hot clique for $K \ge 4$ is never uniform. Interactions with occupied atoms in other polygons also enter the excitation cost, so Eq. (5) is useful but not a sufficient global guarantee.

4. Pulse protocol (DERIVATIONS §6)

Selected schedule. The pulse duration is 5800 ns (200 ns below the device maximum), with $\Omega = 0.75\,\Omega_{\max}$ and fitted δ_f. A three-point Ω ramp (15 % up, 70 % plateau, 15 % down) and a linear detuning sweep during the plateau from $\delta_0 = -\min(\max(|\delta_f|, 2\Omega_{\max}), \delta_{\max})$ to δ_f.

Starting far below resonance puts every atom in $|g\rangle$; the positive final detuning and defect-gap fit target one excitation per customer. Every sequence is built with Pulser on `AnalogDevice` and validated against the device object.

Parameter selection. A scan on three S instances (Fig. 1 right; DERIVATIONS §6.6) was evaluated by P(reach), best gap, and feasibility rather than mean cost. Performance improves with duration at every Ω (2000 → 4000 → 5800 ns); $0.75\,\Omega_{\max}$ exceeds full and half power at 4000 and 5800 ns; $\delta_f \times \{0.8, 1, 1.25\}$ gives 0.915 / 0.932 / 0.884, so the fitted value is retained. S_seed02/03, not used in the scan, reach the model optimum in 99 % / 96 % of samples with these settings. Emulators: pulser-simulation (QuTiP) for every S headline row; emu-sv cross-check on S_seed00 (0.96 vs 0.94); an in-house exact state-vector integrator built from Pulser's own samples for the scans (total-variation distance 0.05–0.07 from QuTiP at 500 shots, i.e. sampling noise); emu-mps for M and L with the bond dimension recorded.

5. Results (brief Table 3; complete table: `results.csv`, 79 rows)

Evaluation metrics. The table reports raw samples without repair. *feas.* denotes the fraction of samples that decode to a capacity-feasible grouping. *model gap*: best feasible shot against the exact model optimum; *CVRP gap*: that shot routed exactly against C^* (proven on S, best known on M/L); both on feasible decodes only, "undefined" if there are none. *P(reach C^*)*: shots whose grouping routes to C^*, so it includes the decomposition loss. *P(gap ≤ 5 %)*: shots within 5 % of the model optimum; on every S instance and on M_seed00/01 this equals P(reach model optimum) because the next-best feasible grouping is 11–62 % worse, on M_seed02 it does not (six groupings within 5 %; DERIVATIONS Q-M3). *fidelity*: relative coupling error of the full matrix, lower is better.

instance	method	feas.	model gap	CVRP gap	P(reach C*)	P(gap≤5%)	fidelity	backend
S_seed00	ours	1.00	0.0 %	0.0 %	0.94	0.94	0.26	qutip
S_seed00	ours, tight A (ablation)	1.00	0.0 %	0.0 %	0.03	0.03	0.31	qutip
S_seed00	random assignments (one-hot)	0.63	0.0 %	0.0 %	0.12	0.12	0.26	random-onehot
S_seed00	qubo-solver quantum	1.00	0.0 %	0.0 %	0.78	0.78	0.12	qs-quantum
S_seed00	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed00	random sampler	0.04	0.0 %	0.0 %	0.01	0.01	0.95	random
S_seed01	ours	0.95	0.0 %	0.0 %	0.95	0.95	0.21	qutip
S_seed02	ours	1.00	0.0 %	0.0 %	0.99	0.99	0.31	qutip
S_seed03	ours	1.00	0.0 %	15.6 %	0.01	0.96	0.29	qutip
S_seed04	ours	1.00	0.0 %	21.7 %	0.02	0.89	0.32	qutip
M_seed00	ours (emu-mps)	0.04	67.9 %	29.9 %	0.00	0.00	0.58	emu-mps χ=32
M_seed00	qubo-solver quantum	0.02	72.1 %	35.2 %	0.00	0.00	0.09	qs-quantum χ=32
M_seed01	ours (emu-mps)	0.39	37.9 %	9.2 %	0.00	0.00	0.08	emu-mps χ=32
M_seed01	qubo-solver quantum	0.37	47.5 %	11.2 %	0.00	0.00	0.08	qs-quantum χ=32
M_seed02	ours (emu-mps)	0.45	23.0 %	16.0 %	0.00	0.00	0.47	emu-mps χ=32
M_seed02	qubo-solver quantum	0.39	34.2 %	13.6 %	0.00	0.00	0.09	qs-quantum χ=32
L_seed00	ours (emu-mps)	0.01	140.0 %	79.7 %	0.00	0.00	0.65	emu-mps χ=32
L_seed01	ours (emu-mps)	0.00	182.9 %	75.1 %	0.00	0.00	0.65	emu-mps χ=32
L_seed02	ours (emu-mps)	0.00	undef.	undef.	0.00	0.00	0.64	emu-mps χ=32

(Source: `scripts/report_tables.py`, generated from `results.csv`. Values are raw samples; repair and additional control rows remain in `results.csv`; the full control block is shown only for S_seed00)

Reference C^* (brief §9.3). CPLEX is not importable, so C^* is the best of qubo-solver's simulated annealing on our penalized QUBO and our constrained annealing on the CVRP, each 10 seeded restarts with a 60 s max solve time (settings in `instances/*.json`). On S the exhaustive dynamic programme is also run and agrees 5/5, so S references are proven; M/L are best known. The QUBO-side annealer reaches C^* on 5/5 S and 2/3 M instances but stays 23–34 % above it on all three L (599.9 / 554.6 / 521.7 vs 447.0 / 434.9 / 424.7). The reported C^* values therefore do not rely exclusively on the QUBO annealer.

instance	ρ	C* (best known)	model optimum routed exactly	decomposition gap
S_seed00	0.714	262.97	262.97	0.0 %
S_seed01	0.857	245.02	245.02	0.0 %
S_seed02	0.714	254.62	254.63	0.0 %
S_seed03	0.643	197.44	228.34	15.6 %
S_seed04	0.714	184.42	224.49	21.7 %
M_seed00	0.950	432.00	432.00	0.0 %
M_seed01	0.733	368.36	368.36	0.0 %
M_seed02	0.700	360.86	385.62	6.9 %
L_seed00	0.650	446.97	525.71	17.6 %
L_seed01	0.708	434.91	470.67	8.2 %
L_seed02	0.600	424.66	480.17	13.1 %

Model selection, evaluated classically. Each additive objective below is solved exactly with capacity enforced and its groups are then routed

exactly. These numbers test objective quality only; the alternatives have not been mapped to the analog register. `depot-MST` means $\mathrm{MST}(S) + 2\min_{i\in S} d_{0i}$ per group.

tier	pairwise routed gaps	medoid routed gaps	depot-MST routed gaps
S	0.0% / 0.0% / 0.0% / 15.6% / 21.7%	6.4% / 0.0% / 3.1% / 0.0% / 17.3%	0.0% / 0.0% / 0.0% / 0.0% / 17.3%
M	0.0% / 0.0% / 6.9%	0.0% / 0.0% / 6.9%	4.4% / 2.6% / 7.4%
L	17.6% / 8.2% / 13.1%	15.6% / 19.8% / 19.8%	0.0% / 0.0% / 7.5%

The depot-aware surrogate removes the observed pairwise-objective loss on L_seed00/01 and cuts it on L_seed02, but worsens every M instance. It is a promising L objective, not a universal replacement.

Capacity-binding validation. From seeds 1000–1199 we selected the first eight S instances for which every unconstrained pairwise optimum overloaded a vehicle. Four development cases selected $\kappa = 0.6$ from $\{0, .05, .15, .3, .6, 1\}$ before the four validation cases were inspected. All values below are raw 500-shot QuTiP runs; intervals are 95 % Wilson intervals conditional on each fixed sampler.

validation instance	feasibility κ=.15	feasibility κ=.6	P(model optimum) .15 / .6	P(C*) .15 / .6
S_seed1052	0.03 [0.02,0.05]	1.00 [0.99,1.00]	0.00 / 0.00	0.00 / 0.05
S_seed1053	0.85 [0.81,0.87]	0.19 [0.16,0.23]	0.84 / 0.01	0.84 / 0.01
S_seed1055	0.04 [0.03,0.06]	1.00 [0.99,1.00]	0.01 / 0.94	0.00 / 0.00
S_seed1057	1.00 [0.99,1.00]	1.00 [0.99,1.00]	0.00 / 0.26	0.70 / 0.71

The stronger weight raises mean feasibility on this deliberately difficult holdout from 0.48 to 0.80, but success is instance-dependent: it hurts S_seed1053 and lowers mean P(reach CVRP reference). It is therefore an ablation, not a new headline default. A bounded classical relocation/swap repair raises M feasibility to 0.98–1.00 and can reach 0 % route gap, but matched random one-hot samples do the same. Repair results remain separately labelled in `out/two_hour/repair/comparison.csv`; raw sampling remains the quantum claim.

Tier S performance. The model-optimal grouping is returned in 89–99 % of raw samples, of which 0.95–1.00 are feasible. Matched random one-hot samples reach the optimum with probability $2/16 \approx 0.12$, while uniform random bitstrings reach it in 0.7–1.2 % of samples. On S_seed03/04 the model gap is zero but P(reach C^*) is 0.01/0.02; the corresponding 15.6/21.7 % loss is therefore attributable to the clustering surrogate rather than sampling.

Assignment-penalty ablation. Replacing A by its tight provable bound (DERIVATIONS Q-A), while retaining the same register construction and pulse, moves P(reach model optimum) from 0.94 to 0.03 on S_seed00, 0.95 to 0.52 on S_seed01 and 0.89 to 0.94 on S_seed04: a smaller A shrinks the defect gaps the sweep must resolve.

Tier M performance. With δ_f bounded by Eq. (4), 98–100 % of samples are one-hot, but only 4.4/38.8/44.6 % are capacity-feasible on M_seed00/01/02. Their best CVRP gaps are 29.9/9.2/16.0 %. Exhaustive analysis of all 59 049 one-hot assignments places the model optimum at realized-device ranks 10 474/12 966/4 192. The capacity term explains 10.2 percentage points of model gap only on M_seed02; geometric distortion is the dominant loss. An independent qubo-solver pipeline exhibits comparable behavior. M_seed00 is converged at $\chi = 32$, whereas M_seed01 and the tier-L simulations reach the bond-dimension cap and remain provisional (DERIVATIONS §6.8).

Tier L performance. L_seed00 / L_seed01 / L_seed02 (60 atoms, emu-mps $\chi = 32$, 500 shots) produce 0.8 % / 0.4 % / 0 % feasible shots; the first two have 79.7 % / 75.1 % CVRP gaps and the third gap is undefined. All three runs hit the bond-dimension cap and are provisional. The weakest square coupling limits δ_f to 4.475 rad/µs and compresses final defect gaps, while the model optimum ranks 16 534 / 1 063 / 14 862 among the one-hot energies on L_seed00/01/02. These diagnostics make embedding and dynamics plausible contributors; they do not measure the minimum many-body spectral gap during the pulse.

6. Limitations and what did not work

- **Embedding accuracy.** Generic placement gives coupling error 0.95 and P(reach) of 3–13 %. One-hot-subspace fitting raises tier-S P(reach) to 0.89–0.99, but no tested planar register preserves the intended ordering for $K \geq 3$.
- **Detuning bounds.** Unbounded tier-M detuning favors multiple excitations per customer and yields no decodable samples on two instances. The weakest intra-polygon coupling therefore bounds δ_f in every reported run.
- **Penalty and register sensitivity.** The capacity coefficient shifts the M_seed02 QUBO optimum by 10.2 %, and overlapping optimizer runs produced different registers from the same seed. Reported experiments use frozen weights and committed registers; the held-out capacity study is presented separately.
- **Classical post-processing.** Capacity-local repair frequently produces feasible, short routes, but matched random controls improve by the same amount. Repaired results therefore do not support a quantum advantage.